

MATH 239 Introduction to Combinatorics

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1 Principles of Enumeration

Lecture 1, 2025/09/03

1.1 AND vs. OR - Products vs. Sums

Example. Let $A = \{\text{apple, orange, banana, grape}\}$ and $B = \{\text{carrot, lettuce, onion}\}$. The # of ways of choosing a fruit from A **AND** a vegetable from B is $|A| \cdot |B| = 4 \cdot 3 = 12$. This is equivalent to choosing one element from the cartesian product of A and B :

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

For finite sets, the cardinalities satisfy

$$|A \times B| = |A| \cdot |B|.$$

The # of ways of choosing a fruit from A **OR** a vegetable from B is $|A| + |B| = 4 + 3 = 7$. This is equivalent to choosing one element from the union of A and B :

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

For disjoint sets,

$$|A \cup B| = |A| + |B|$$

Note that not all sets are disjoint, some have non-trivial intersection.

The intersection is $A \cap B = \{x : x \in A \text{ and } x \in B\}$. For finite sets,

$$|A \cup B| + |A \cap B| = |A| + |B|.$$

1.2 Lists, Permutations, and Subsets

Definition 1.1 (List).

A **list** of a set S is a listing of the elements of S exactly once, in some order.

Definition 1.2 (Permutation).

A **permutation** is a list of the set $\{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$.

Example. $S = \{a, b, c\}$ has 6 lists:

$$abc \quad acb \quad bac \quad bca \quad cab \quad cba.$$

We can construct a list of S by choosing an element $v \in S$ to be the first element of the list, and then append a list of $S \setminus \{v\}$. If p_n is the # of lists of a set of size n , for $n \in \mathbb{N}$, then

$$p_n = n \cdot p_{n-1}$$

and $p_1 = 1$. So by induction we have the following theorem.

Theorem 1.1. For every $n \geq 1$, the # of lists of an n -element set is $n \cdot (n-1) \cdots 2 \cdot 1 = n!$.

Note. By convention, we take $0! = 1$.

Definition 1.3 (Subset).

A **subset** of S is a set containing some (perhaps none, or perhaps all) of the elements of S , at most once each, and in no particular order.

Note. There are 2^n subsets of an n -element set.

To specify: a subset X of S , for each element $v \in S$, we have to decide whether v is in X or not in X . Equivalently, we have to choose {in, out} n times. So,

$$\underbrace{|\{\text{in, out}\}| \cdot |\{\text{in, out}\}| \cdot \dots \cdot |\{\text{in, out}\}|}_{n \text{ times}} = 2^n.$$

Lecture 2, 2025/09/05

Definition 1.4 (Partial List).

A **partial list** of a set S is a list of a subset of S .

We are interested in k -element partial lists of an n -element set, in other words, k -tuples $(s_1, \dots, s_k)$ of distinct elements of S . There are n choices for s_1 , $n - 1$ choices for s_2 , ..., $(n - k + 1)$ choices for s_k . So we have the following.

Theorem 1.2. For $k, n \geq 0$, the # of partial lists of length k of an n -element set is

$$n(n-1)(n-2) \cdots (n-k+1).$$

Note. For $0 \leq k \leq n$, we can write

$$n(n-1)(n-2) \cdots (n-k+1) = \frac{n(n-1) \cdots 1}{(n-k)(n-k-1) \cdots 1} = \frac{n!}{(n-k)!}.$$

Definition 1.5. For $n, k \geq 0$, let $\binom{n}{k}$ (read “ n choose k ”) denote the # of k -element subsets of an n -element set.

Example.

- $\binom{n}{0} = 1$.
- $\binom{n}{n} = 1$ for all n .
- $\binom{n}{1} = n$ for all n .
- $\binom{n}{k} = 0$ if $k > n$.

Theorem 1.3. For $0 \leq k \leq n$, we have

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Proof. Let S be a set of size n . From Theorem 1.2, the # of partial lists of S of length k is $\frac{n!}{(n-k)!}$. We will choose a k -element subset X of S AND choose a list of X , which will give a list of a subset of S of length k . Thus, the # of length- k partial lists of S is

$$(\# \text{ of } k\text{-element subsets of } S) \cdot (\# \text{ of lists of a } k\text{-element set}) = \binom{n}{k} \cdot k!.$$

$$\text{So, } \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

□

Note. This is an example of a combinatorial proof, where we prove an identity by counting the same set in two different ways.

Example. For $n \geq 0$,

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

Prove this by counting sets, not by algebraic manipulation.

Proof. The RHS, 2^n , is the # of subsets of an n -element set. On the LHS, for $0 \leq k \leq n$, $\binom{n}{k}$ is the # of k -element subsets of an n -element set. And addition corresponds to “OR”, so the LHS is the # of ways of choosing a single subset of an n -element set. That is, choosing one of the $\binom{n}{0}$ 0-element subsets, or one of the $\binom{n}{1}$ 1-element subsets, or ..., or one of the $\binom{n}{n}$ n -element subsets. $\square$

Example. For $k, n \geq 1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. The LHS, $\binom{n}{k}$, is the # of k -element subsets of an n -element set S . WLOG, assume $S = \{1, 2, \dots, n\}$. Notice that $\binom{n-1}{k}$ is the # of k -element subsets of $S' = \{1, 2, \dots, n-1\}$, and $\binom{n-1}{k-1}$ is the # of $(k-1)$ -element subsets of S' . Recall our intuition that addition corresponds to “OR”. Any k -element subset of S either includes n or does not include n . If $n \notin X$, then X is a subset of size k of S' . If $n \in X$, then $X \setminus \{n\}$ is a subset of size $k-1$ of S' . Thus, LHS = RHS. $\square$

1.3 Multisets

Suppose I have a bag containing 8 marbles, each of which is either red, green, or blue. How many different bag contents are possible?

First question: how can I represent the contents of a bag? For example, we could have 2 red, 3 green, and 3 blue marbles.

Lecture 3, 2025/09/8

- We might want to write it as a set $\{R, R, G, G, B, B, B\}$, but this is not a set since sets do not have duplicates.
- Invent new notation that preserves multiplicity:

$$[[R, R, G, G, G, B, B, B]] \text{ or } \langle\langle R^2, G^3, B^3 \rangle\rangle.$$

- Keep track of multiplicities in a vector. Fix an order, say, R then G then B , and use a tuple or a vector to record the multiplicities: $(2, 3, 3)$.

To count the # of bags containing 8 marbles each of which is either red, green, or blue, we need to count the # of 3-tuples (n_1, n_2, n_3) of integers s.t. $n_1 + n_2 + n_3 = 8$ and $n_1, n_2, n_3 \geq 0$.

Definition 1.6 (Multiset).

Let $n \geq 0, t \geq 1$ be integers. A **multiset** of size n with elements of t types is a sequence of non-negative integers $m_1, m_2, \dots, m_t$ s.t. $m_1 + m_2 + \dots + m_t = n$. The set of all multisets of size n with elements of t -types is denoted $\mathcal{M}(n, t)$.

Theorem 1.4. For $n \geq 0, t \geq 1$, the # of n -element multisets with elements of t types is

$$|\mathcal{M}(n, t)| = \binom{n+t-1}{t-1}.$$

Proof. We know that $\binom{n+t-1}{t-1}$ is the # of $(t-1)$ -element subsets of an $(n+t-1)$ -element set. We will show how to translate between $(t-1)$ -element subsets of an $(n+t-1)$ -element set and multisets of size n with t types. Let us work down a row of $n+t-1$ circles (e.g. $n=8, t=3$):

○ ○ ○ ○ ○ ○ ○ ○ ○ ○

Now, let us choose a $(t-1)$ -element subset of these and cross them out

○ ○ × ○ ○ ○ × ○ ○ ○

Our row now has n circles grouped into t segments each containing 0 or more consecutive circles. Let m_i be the length of the i^{th} segment of consecutive circles, so $m_1 + m_2 + \dots + m_t = n$. So $(m_1, m_2, \dots, m_t)$ is a multiset of size n with elements of t types.

Now, let us go the other way. Let $(m_1, m_2, \dots, m_t)$ be a multiset of size n with t types. Write down m_1 circles, then an $\times$, then m_2 circles, then an $\times$, and so on, upto m_t circles. We will have written down $n+t-1$ symbols total, of which $t-1$ are $\times$'s, these indicate the $(t-1)$ -element subset of an $(n+t-1)$ -element set. □

Note. This is an example of a bijective proof.

1.4 Bijections

Definition 1.7 (Surjective, Injective, Bijective).

Let A and B be sets. Let $f : A \rightarrow B$.

- f is **surjective** (or **onto**) if, for every $b \in B$, $\exists a \in A$ s.t. $f(a) = b$.
- f is **injective** (or **one-to-one**) if, for every $a, a' \in A$, if $f(a) = f(a')$, then $a = a'$.
- f is **bijective** if it is both surjective and injective.

Definition 1.8 (Well-defined).

An expression yields a **well-defined** function $f : A \rightarrow B$ if, for every input $a \in A$, $f(a)$ assigns a single unambiguous value $b \in B$.

Proposition 1.5. A function $f : A \rightarrow B$ is a bijection $\iff f$ has an inverse, namely if $\exists$ a well-defined function $g : B \rightarrow A$ s.t. $g(f(a)) = a$ for all $a \in A$ and $f(g(b)) = b$ for all $b \in B$.

Remark (Notation).

If $\exists$ a bijection between sets A and B , we can write $A \rightleftharpoons B$.

Corollary 1.6. If $A \rightleftharpoons B$ and at least one of A or B is finite, then $|A| = |B|$.

Example. For $k, n \geq 1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. Let $S = \{1, 2, \dots, n\}$ and $S' = \{1, 2, \dots, n-1\}$. Let $A = \{k\text{-element subsets of } S\}$ and $B = \{(k-1)\text{-element or } k\text{-element subsets of } S'\}$. By definition, $|A| = \binom{n}{k}$ and we can easily see that $|B| = \binom{n-1}{k-1} + \binom{n-1}{k}$. The goal is to prove $A \rightleftharpoons B$. Define $f : A \rightarrow B$ by

$$f(u) = \begin{cases} u, & \text{if } n \notin u \\ u \setminus \{n\}, & \text{if } n \in u \end{cases}$$

Lecture 4, 2025/09/10

Question: Is f well-defined, i.e. $\forall u \in A$, is $f(u) \in B$?

Answer: Yes, $f(u)$ does not contain n (so it's a subset of S') and has size either $k-1$ or k . So

$f(u) \in B$.

Define $g : B \rightarrow A$ by

$$g(v) = \begin{cases} v, & \text{if } |v| = k \\ v \cup \{n\}, & \text{if } |v| = k - 1 \end{cases}$$

Claim (1). For all $u \in A$, $g(f(u)) = u$.

Proof of Claim (1). First consider $u \in A$ s.t. $n \notin u$. Then, $f(u) = u$ and $|f(u)| = k$, so $g(f(u)) = u$ (first branch of g). Now consider $u \in A$ s.t. $n \in u$. Then, $f(u) = u \setminus \{n\}$ and $|f(u)| = k - 1$, so $g(f(u)) = (u \setminus \{n\}) \cup \{n\} = u$ (second branch of g). $\square$

Claim (2). For all $v \in B$, $f(g(v)) = v$.

Proof of Claim (2). Exercise. $\square$

Thus, f is a bijection, so $|A| = |B|$, i.e. $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. $\square$

Remark (Problem Solving Tips: what steps are needed for a bijective argument?).

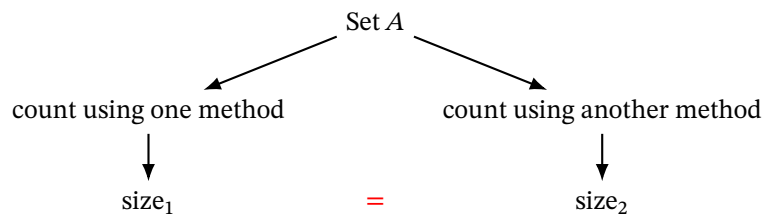
To show a function $f : A \rightarrow B$ is a bijection by giving an inverse:

- (1) State the function $f : A \rightarrow B$.
- (2) Show that f is well-defined, i.e. $\forall a \in A, f(a) \in B$.
- (3) State the function $g : B \rightarrow A$.
- (4) Show that g is well-defined.
- (5) Show that $g(f(a)) = a$ for all $a \in A$.
- (6) Show that $f(g(b)) = b$ for all $b \in B$.

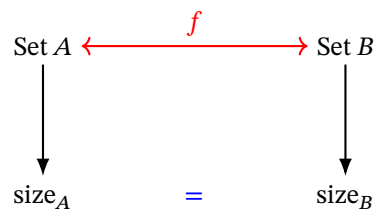
Exercise: Show that $\binom{n}{k} = \binom{n}{n-k}$ for $n \geq k \geq 0$ using a combinatorial or bijective argument.

1.5 Summary of Combinatorial and Bijective Proofs

In a combinatorial proof, describe two different ways of counting the size of the same set.



In a bijective proof, describe two different sets and show they're in bijection.



- (1) Give sets A, B and count them.
- (2) Show a bijection between A and B .
- (3) This implies $|A| = |B|$.

2 Generating Series

Definition 2.1 (Formal Power Series).

A **formal power series** is an expression of the form

$$G(x) = \sum_{n=0}^{\infty} g_n x^n$$

in which the coefficients $(g_0, g_1, g_2, \dots)$ are a sequence of real numbers.

Note. We use x as an **indeterminate** for which we do not normally substitute in particular values. This means that we never worry about the convergence of the series, all we require is that the coefficients are finite real numbers.

Example. $G(x) = 0 - 1x + 2x^2 - 3x^3 + \dots = \sum_{n=0}^{\infty} (-1)^n n x^n$.

We can add and multiply formal power series like polynomials. Let

$$F(x) = \sum_{n \geq 0} f_n x^n = f_0 + f_1 x + f_2 x^2 + \dots,$$

$$G(x) = \sum_{n \geq 0} g_n x^n = g_0 + g_1 x + g_2 x^2 + \dots.$$

Then,

$$(F + G)(x) = \sum_{n \geq 0} (f_n + g_n) x^n = (f_0 + g_0) + (f_1 + g_1)x + (f_2 + g_2)x^2 + \dots,$$

$$(F \cdot G)(x) = \sum_{n \geq 0} \left(\sum_{k=0}^n f_k g_{n-k} \right) x^n = (f_0 g_0) + (f_0 g_1 + f_1 g_0)x + (f_0 g_2 + f_1 g_1 + f_2 g_0)x^2 + \dots.$$

Lecture 5, 2025/09/12

Example. Note that

$$\begin{aligned} (1 + 3x^2 + 5x^4 + \dots) + x(2 + 4x^2 + 6x^4 + \dots) &= (1 + 3x^2 + 5x^4 + \dots) + (2x + 4x^3 + 6x^5 + \dots) \\ &= 1 + 2x + 3x^2 + 4x^3 + 5x^4 + 6x^5 + \dots \end{aligned}$$

Equivalently,

$$\begin{aligned}\sum_{n \geq 0} (2n+1)x^{2n} + x \sum_{n \geq 0} (2n+2)x^{2n} &= \sum_{n \geq 0} (2n+1)x^{2n} + \sum_{n \geq 0} (2n+2)x^{2n+1} \\ &= \sum_{n \geq 0} ((2n+1)x^{2n} + (2n+2)x^{2n+1}) \\ &= \sum_{n \geq 0} (n+1)x^n.\end{aligned}$$

Definition 2.2 (Geometric Series).

The **geometric series** is of the form

$$G(x) = 1 + x + x^2 + x^3 + \cdots = \sum_{n \geq 0} x^n.$$

Consider $xG(x) = x + x^2 + x^3 + \cdots = x \sum_{n \geq 0} x^n$. Then,

$$G(x) - xG(x) = 1 \implies G(x)(1-x) = 1 \implies G(x) = \frac{1}{1-x} = (1-x)^{-1}.$$

Definition 2.3 (Inverse of a Formal Power Series).

If $F(x)$ and $G(x)$ are formal power series such that $F(x)G(x) = 1$, then we write $G(x) = F(x)^{-1}$ or $G(x) = \frac{1}{F(x)}$, and say that $G(x)$ is the **inverse** of $F(x)$.

Corollary 2.1. The inverse of $F(x) = \sum_{n \geq 0} f_n x^n$ exists $\iff f_0 \neq 0$.

2.1 Binomial Theorem

Theorem 2.2 (Binomial Theorem).

For any $n \in \{0, 1, 2, \dots\}$,

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k = \sum_{k \geq 0} \binom{n}{k} x^k.$$

Proof. We will use a combinatorial proof. Let $\mathcal{P}_n$ be the power set of $\{1, 2, \dots, n\}$, i.e. the set of all subsets of $\{1, 2, \dots, n\}$. Let $\mathcal{P}_{n,k}$ be the set of all k -element subsets of $\{1, 2, \dots, n\}$. We saw before that a

bijection between $\mathcal{P}_n$ and $\{0, 1\}^n$, i.e. a subset S of $\{1, 2, \dots, n\}$ corresponds to an “indicator vector” in $\{0, 1\}^n$. Let $S \in \mathcal{P}_n$ be a subset of $\{1, 2, \dots, n\}$. The corresponding indicator vector will be

$$\vec{a} = (a_1, a_2, \dots, a_n) \quad \text{where} \quad a_i = \begin{cases} 0, & \text{if } i \notin S, \\ 1, & \text{if } i \in S. \end{cases}$$

Note that $|S| = \sum_{i=1}^n a_i$. Let’s introduce an indeterminate x . For every subset S with indicator vector $\vec{a}$, it holds that

$$x^{|S|} = x^{a_1+a_2+\dots+a_n}.$$

Because of the bijection between $\mathcal{P}_n$ and $\{0, 1\}^n$, summing over all subsets is equivalent to summing over all indicator vectors. Thus,

$$\sum_{S \in \mathcal{P}_n} x^{|S|} = \sum_{\vec{a} \in \{0,1\}^n} x^{a_1+a_2+\dots+a_n}.$$

Now, let’s simplify the equation.

$$\begin{aligned} \text{LHS} &= \sum_{S \in \mathcal{P}_n} x^{|S|} = \sum_{k=0}^n \sum_{S \in \mathcal{P}_{n,k}} x^{|S|} && \text{since } \mathcal{P}_n = \bigcup_{k=0}^n \mathcal{P}_{n,k} \\ &= \sum_{k=0}^n \sum_{S \in \mathcal{P}_{n,k}} x^k && \text{since } |S| = k \quad \forall S \in \mathcal{P}_{n,k} \\ &= \sum_{k=0}^n x^k |\mathcal{P}_{n,k}| && \text{since } x^k \text{ does not depend on } S \\ &= \sum_{k=0}^n \binom{n}{k} x^k && \text{since } |\mathcal{P}_{n,k}| = \binom{n}{k}. \end{aligned}$$

On the RHS, summing over all $\vec{a} \in \{0, 1\}^n$ can be broken up into summing over all $a_1 \in \{0, 1\}$,

$a_2 \in \{0, 1\}, \dots, a_n \in \{0, 1\}$. Thus,

$$\begin{aligned}
\text{RHS} &= \sum_{\vec{a} \in \{0,1\}^n} x^{a_1+a_2+\dots+a_n} = \sum_{(a_1, a_2, \dots, a_n) \in \{0,1\}^n} x^{a_1} x^{a_2} \dots x^{a_n} && \text{(Cartesian product)} \\
&= \sum_{a_1 \in \{0,1\}} \sum_{a_2 \in \{0,1\}} \dots \sum_{a_n \in \{0,1\}} x^{a_1} x^{a_2} \dots x^{a_n} \\
&= \sum_{a_1=0}^1 x^{a_1} \sum_{a_2=0}^1 x^{a_2} \dots \sum_{a_n=0}^1 x^{a_n} \\
&= \left(\sum_{a_1=0}^1 x^{a_1} \right) \left(\sum_{a_2=0}^1 x^{a_2} \right) \dots \left(\sum_{a_n=0}^1 x^{a_n} \right) \\
&= \underbrace{(1+x)(1+x) \dots (1+x)}_{n \text{ times}} \\
&= (1+x)^n. \quad \square
\end{aligned}$$

Lecture 6, 2025/09/15

Theorem 2.3 (Negative Binomial Theorem).

If $t \geq 1$, then

$$(1-x)^{-t} = \sum_{n \geq 0} \binom{n+t-1}{t-1} x^n.$$

Proof. We will use a combinatorial proof. We recognize that $\binom{n+t-1}{t-1}$ is $|\mathcal{M}(n, t)|$, the # of multisets $(m_1, \dots, m_t)$ of size n with elements of t types. Let $\mathcal{M}(t)$ be the set of all multisets (with any # of elements) of t types. Observe that

$$\mathcal{M}(t) = \bigcup_{n \geq 0} \mathcal{M}(n, t).$$

Note that an element of $\mathcal{M}(t)$ is a tuple $(m_1, \dots, m_t)$ with $m_i \in \mathbb{N}$, so $\mathcal{M}(t) = \mathbb{N}^t$. We will start from

the RHS.

$$\begin{aligned}
\sum_{n \geq 0} \binom{n+t-1}{t-1} x^n &= \sum_{n \geq 0} |\mathcal{M}(n, t)| x^n \\
&= \sum_{n \geq 0} \left(\sum_{\mu \in \mathcal{M}(n, t)} 1 \right) x^n \\
&= \sum_{n \geq 0} \sum_{\mu \in \mathcal{M}(n, t)} x^n \\
&= \sum_{n \geq 0} \sum_{\mu \in \mathcal{M}(t)} x^{|\mu|} && \text{since } n = |\mu| \\
&= \sum_{\mu \in \mathcal{M}(t)} x^{|\mu|} && \text{since } \mathcal{M}(t) = \bigcup_{n \geq 0} \mathcal{M}(n, t) \\
&= \sum_{(m_1, \dots, m_t) \in \mathbb{N}^t} x^{m_1 + m_2 + \dots + m_t} && \text{since } \mathcal{M}(t) = \mathbb{N}^t \\
&= \sum_{m_1 \geq 0} \sum_{m_2 \geq 0} \dots \sum_{m_t \geq 0} x^{m_1 + m_2 + \dots + m_t} \\
&= \sum_{m_1 \geq 0} x^{m_1} \sum_{m_2 \geq 0} x^{m_2} \dots \sum_{m_t \geq 0} x^{m_t} \\
&= \left(\sum_{m_1 \geq 0} x^{m_1} \right) \left(\sum_{m_2 \geq 0} x^{m_2} \right) \dots \left(\sum_{m_t \geq 0} x^{m_t} \right) \\
&= \frac{1}{1-x} \cdot \frac{1}{1-x} \dots \frac{1}{1-x} \\
&= \left(\frac{1}{1-x} \right)^t. \quad \square
\end{aligned}$$

Example. Write $\left(\frac{x^2}{1+3x} \right)^4$ as a formal power series.

Proof. Note that

$$\begin{aligned}
\left(\frac{x^2}{1+3x} \right)^4 &= x^8 \left(\frac{1}{1-(-3x)} \right)^4 \\
&= x^8 \sum_{n \geq 0} \binom{n+4-1}{4-1} (-3x)^n \\
&= \sum_{n \geq 0} \binom{n+3}{3} (-3)^n x^{n+8} \\
&= \sum_{n \geq 8} \binom{n-5}{3} (-3)^{n-8} x^n. \quad \square
\end{aligned}$$

Definition 2.4 (Coefficient Extraction).

Let $G(x) = \sum_{n \geq 0} g_n x^n$ be a formal power series. Let $k \in \mathbb{N}$, define

$$[x^k]G(x) = g_k$$

i.e. $[x^k]$ is an operator that extracts the coefficient of x^k from $G(x)$.

Example (Continued).

We have

$$[x^k] \left(\frac{x^2}{1+3x} \right)^4 = [x^k] \sum_{n \geq 8} \binom{n-5}{3} (-3)^{n-8} x^n = \begin{cases} \binom{k-5}{3} (-3)^{k-8}, & k \geq 8, \\ 0, & 0 \leq k < 8. \end{cases}$$

Remark (Some Rules about Coefficient Extraction).

- (1) $[x^k](aF(x) + bG(x)) = a[x^k]F(x) + b[x^k]G(x)$ for $a, b \in \mathbb{R}$.
- (2) $[x^k](x^\ell F(x)) = [x^{k-\ell}]F(x)$ for $\ell \in \mathbb{Z}$.
- (3) $[x^k](F(x)G(x)) = \sum_{\ell=0}^k ([x^\ell]F(x)) \cdot ([x^{k-\ell}]G(x))$.

2.2 Generating Series

We will use formal power series to record counting information about a set, in which case we call it a **generating series**.

Example. Let $\mathcal{M} = \{\text{Jan, Feb, } \dots, \text{Dec}\}$. Let

$$\mathcal{M}_n = \{\alpha \in \mathcal{M} : \alpha \text{ has exactly } n \text{ days (no leap year)}\}.$$

Note that $\mathcal{M} = \bigcup_{n \geq 0} \mathcal{M}_n$. For example,

$$\mathcal{M}_0 = \emptyset, \quad \mathcal{M}_{28} = \{\text{Feb}\}.$$

Consider

$$\sum_{n \geq 0} |\mathcal{M}_n| x^n = x^{28} + 4x^{30} + 7x^{31}.$$

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We could derive the same summation in a different way. Consider the following.

$$\sum_{\alpha \in \mathcal{M}} x^{\# \text{ of days in } \alpha} = x^{31} + x^{28} + x^{31} + x^{30} + \cdots + x^{31} = x^{28} + 4x^{30} + 7x^{31}.$$

Definition 2.5 (Weight Function).

Let S be a set. A **weight function** is a function $\omega : S \rightarrow \mathbb{N}$ s.t. for every $n \in \mathbb{N}$, the # of elements of S of weight n is finite, i.e. $\{\alpha \in S : \omega(\alpha) = n\}$ is finite.

Definition 2.6 (Generating Series).

Let S be a set and ω be a weight function on S . The **generating series** of S with respect to ω is

$$\Phi_S^\omega(x) = \Phi_S(x) = \sum_{\alpha \in S} x^{\omega(\alpha)}.$$

Note. Coefficients in a generating series are always non-negative integers.

In our example above,

$$\Phi_{\mathcal{M}}^{\# \text{ of days}}(x) = \sum_{\alpha \in \mathcal{M}} x^{\# \text{ of days in } \alpha} = x^{28} + 4x^{30} + 7x^{31}.$$

Proposition 2.4. Let ω be a weight function on a set S . Then,

$$\Phi_S^\omega(x) = \sum_{n \geq 0} \underbrace{|\{\alpha \in S : \omega(\alpha) = n\}|}_{\# \text{ of elements of weight } n \text{ in } S} x^n.$$

Equivalently,

$$[x^k] \Phi_S^\omega(x) = |\{\alpha \in S : \omega(\alpha) = k\}|.$$

Proof. Let $S_n = \{\alpha \in S : \omega(\alpha) = n\}$. Note that $S = \bigcup_{n \geq 0} S_n$. Then,

$$\Phi_S^\omega(x) = \sum_{\alpha \in S} x^{\omega(\alpha)} = \sum_{n \geq 0} \sum_{\alpha \in S_n} x^{\omega(\alpha)} = \sum_{n \geq 0} \sum_{\alpha \in S_n} x^n = \sum_{n \geq 0} |S_n| x^n.$$

□

Example. Write the generating series for the set S of all binary strings with respect to the weight function ω which records the length of the string.

Proof. We have

$$\begin{aligned} \Phi_S^\omega(x) &= \sum_{\alpha \in S} x^{\omega(\alpha)} \\ &= \sum_{n \geq 0} |\{\alpha \in S : \omega(\alpha) = n\}| x^n = \sum_{n \geq 0} (\# \text{ of binary strings of length } n) x^n \\ &= \sum_{n \geq 0} 2^n x^n \\ &= \sum_{n \geq 0} (2x)^n = \frac{1}{1 - 2x}. \end{aligned}$$

□

Example. Write the generating series for the set

$$S = \{\text{subsets of } \{1, 2, \dots, t\}\}$$

with the weight function $\omega(\alpha) = |\alpha|$.

Proof. We have

$$\begin{aligned} \Phi_S^\omega(x) &= \sum_{\alpha \subseteq \{1, 2, \dots, t\}} x^{|\alpha|} \\ &= \sum_{n \geq 0} (\# \text{ of subsets of size } n) x^n \\ &= \sum_{n \geq 0} \binom{t}{n} x^n \\ &= (1 + x)^t. \end{aligned}$$

□

2.3 Sum and Product Lemmas

Example. Consider breakfast foods and a weight function that corresponds to their cost. Let

$$S_1 = \{\text{omelette, waffles, pancakes, eggs, cereal}\},$$

and $\omega_1 : S_1 \rightarrow \mathbb{N}$ be defined as follows:

$$\omega_1(\text{omelette}) = 10, \quad \omega_1(\text{waffles}) = 10, \quad \omega_1(\text{pancakes}) = 8, \quad \omega_1(\text{eggs}) = 8, \quad \omega_1(\text{cereal}) = 5.$$

The generating series for breakfast main dishes with respect to cost is

$$\Phi_{S_1}^{\omega_1}(x) = \sum_{\alpha \in S_1} x^{\omega_1(\alpha)} = x^{10} + x^{10} + x^8 + x^8 + x^5 = 2x^{10} + 2x^8 + x^5.$$

Equivalently,

$$\Phi_{S_1}^{\omega_1}(x) = \sum_{n \geq 0} |\{\alpha \in S_1 : \omega_1(\alpha) = n\}| x^n = 2x^{10} + 2x^8 + x^5,$$

where the exponents are the costs and the coefficients are the # of things of that cost.

Let

$$S_2 = \{\text{bacon, hashbrowns, toast}\},$$

and $\omega_2 : S_2 \rightarrow \mathbb{N}$ be defined as follows:

$$\omega_2(\text{bacon}) = 5, \quad \omega_2(\text{hashbrowns}) = 4, \quad \omega_2(\text{toast}) = 3.$$

So the generating series for S_2 with respect to cost is

$$\Phi_{S_2}^{\omega_2}(x) = \sum_{\alpha \in S_2} x^{\omega_2(\alpha)} = x^5 + x^4 + x^3.$$

Now, let's consider

$$S_1 \cup S_2 = \{\text{omelette, waffles, pancakes, eggs, cereal, bacon, hashbrowns, toast}\}.$$

The generating series for any breakfast food (main or side) with respect to cost is

$$\Phi_{S_1 \cup S_2}^\omega(x) = \sum_{\alpha \in S_1 \cup S_2} x^{\omega(\alpha)} = 2x^{10} + 2x^8 + x^5 + x^5 + x^4 + x^3 = 2x^{10} + 2x^8 + 2x^5 + x^4 + x^3,$$

Observe that $\Phi_{S_1 \cup S_2}^\omega(x) = \Phi_{S_1}^{\omega_1}(x) + \Phi_{S_2}^{\omega_2}(x)$.

Note. In this case, S_1 and S_2 are disjoint.

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Now, consider

$$\begin{aligned} \Phi_{S_1}^{\omega_1}(x) \cdot \Phi_{S_2}^{\omega_2}(x) &= (2x^{10} + 2x^8 + x^5)(x^5 + x^4 + x^3) \\ &= 2x^{15} + 2x^{14} + 2x^{13} + 2x^{13} + 2x^{12} + 2x^{11} + x^{10} + x^9 + x^8 \\ &= 2x^{15} + 2x^{14} + 4x^{13} + 2x^{12} + 2x^{11} + x^{10} + x^9 + x^8. \end{aligned}$$

For example, $2x^{12}$ indicates that there are two (main, side) combos with total price 12. So this is also the generating series for the cartesian product $S_1 \times S_2$ of breakfast combos with respect to cost. This can be denoted as

$$\Phi_{S_1 \times S_2}^{\omega'}(x)$$

where $\omega'(\alpha_1, \alpha_2) = \omega_1(\alpha_1) + \omega_2(\alpha_2)$.

Lemma 2.5 (Sum Lemma).

Let S_1 and S_2 be disjoint sets and let ω be a weight function on $S_1 \cup S_2$. Then,

$$\Phi_{S_1 \cup S_2}^\omega(x) = \Phi_{S_1}^\omega(x) + \Phi_{S_2}^\omega(x).$$

Proof. The main idea is that sums correspond to disjoint unions.

$$\begin{aligned} \Phi_{S_1}(x) + \Phi_{S_2}(x) &= \sum_{\alpha \in S_1} x^{\omega(\alpha)} + \sum_{\alpha \in S_2} x^{\omega(\alpha)} \\ &= \sum_{\alpha \in S_1 \cup S_2} x^{\omega(\alpha)} && \text{since } S_1 \text{ and } S_2 \text{ are disjoint} \\ &= \Phi_{S_1 \cup S_2}^\omega(x). && \square \end{aligned}$$

Lemma 2.6 (Infinite Sum Lemma).

Let $S_0, S_1, S_2, \dots$ be disjoint sets with union S and let ω be a weight function on S . Then,

$$\Phi_S^\omega(x) = \sum_{n \geq 0} \Phi_{S_n}^\omega(x).$$

Proof. Similar to the proof of the Sum Lemma. □

Lemma 2.7 (Product Lemma).

Let S_1 and S_2 be sets and let ω_1 and ω_2 be weight functions on S_1 and S_2 respectively. Then,

$$\Phi_{S_1 \times S_2}^\omega(x) = \Phi_{S_1}^{\omega_1}(x) \cdot \Phi_{S_2}^{\omega_2}(x)$$

where ω is a weight function on $S_1 \times S_2$ defined by

$$\omega(\alpha, \beta) = \omega_1(\alpha) + \omega_2(\beta).$$

Proof. The main idea is that products correspond to cartesian products.

$$\begin{aligned} \Phi_{S_1}^{\omega_1}(x) \cdot \Phi_{S_2}^{\omega_2}(x) &= \sum_{\alpha_1 \in S_1} x^{\omega_1(\alpha_1)} \cdot \sum_{\alpha_2 \in S_2} x^{\omega_2(\alpha_2)} \\ &= \sum_{\alpha_1 \in S_1} \sum_{\alpha_2 \in S_2} x^{\omega_1(\alpha_1) + \omega_2(\alpha_2)} \\ &= \sum_{(\alpha_1, \alpha_2) \in S_1 \times S_2} x^{\omega(\alpha_1, \alpha_2)} \\ &= \Phi_{S_1 \times S_2}^\omega(x). \end{aligned}$$

□

Example. Let $S = \{2, 4, 6, \dots\}$. How many pairs (a, b) are there with $a, b \in S$, and $a + b = 50$?

Proof. Let $\omega : S \rightarrow \mathbb{N}$ be defined by $\omega(\alpha) = \alpha$ and let $\omega' : S \times S \rightarrow \mathbb{N}$ be defined by

$\omega'(\alpha, \beta) = \alpha + \beta$. Note that we are looking for $[x^{50}]\Phi_{S \times S}^{\omega'}(x)$.

$$\begin{aligned}
[x^{50}]\Phi_{S \times S}^{\omega'}(x) &= [x^{50}](\Phi_S^{\omega}(x))^2 && \text{by Product Lemma} \\
&= [x^{50}]\left(\sum_{\alpha \in S} x^{\omega(\alpha)}\right)^2 \\
&= [x^{50}](x^2 + x^4 + x^6 + \dots)^2 \\
&= [x^{50}](x^2(1 + x^2 + x^4 + \dots))^2 \\
&= [x^{50}]\frac{x^4}{(1 - x^2)^2} \\
&= [x^{46}]\frac{1}{(1 - x^2)^2} && \text{by coefficient extraction rule (2)} \\
&= [x^{46}]\sum_{n \geq 0} \binom{n+2-1}{2-1}(x^2)^n && \text{by Negative Binomial Theorem} \\
&= [x^{46}]\sum_{n \geq 0} (n+1)x^{2n} \\
&= 23 + 1 && \text{taking } n = 23 \\
&= 24. && \square
\end{aligned}$$

Definition 2.7 (Star Operator).

Let A be a set. Define

$$A^* = \bigcup_{k \geq 0} A^k = \{\text{all tuples (of any length) of elements of } A\}.$$

Example. $\{0, 1\}^* = \{\underbrace{()}_{A^0}, \underbrace{(\overbrace{0}, \overbrace{1})}_{A^1}, \underbrace{(\overbrace{0, 0}, \overbrace{0, 1}, \overbrace{1, 0}, \overbrace{1, 1})}_{A^2}, \underbrace{(\overbrace{0, 0, 0}, \dots)}_{A^3, \dots}\}.$

Example. $\{1, 2, 3, \dots\}^* = \{\underbrace{()}_{A^0}, \underbrace{(1), (2), (3), \dots}_{A^1}, \underbrace{(1, 1), (1, 2), \dots}_{A^2}, \underbrace{(1, 1, 1), \dots}_{A^3, \dots}\}.$

Given a weight function ω on A , we can define ω^* on A^* such that

$$\omega^*((\alpha_1, \dots, \alpha_k)) = \sum_{i=1}^k \omega(\alpha_i).$$

This is a well-defined weight function provided no elements of A have weight 0 (otherwise, there would be infinitely many tuples of any given weight).

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Example. Let $S = \{0, 1\}$ and let ω count length i.e. $\omega(0) = \omega(1) = 1$. Then,

$$\omega^*((1, 1, 0, 1)) = \omega(1) + \omega(1) + \omega(0) + \omega(1) = 4$$

$$\omega^*((1)) = \omega(1) = 1$$

$$\omega^*() = 0.$$

So ω^* also counts length of tuples. Recall that the definition of ω^* requires no elements to have weight 0 under ω . Why?

Suppose γ had weight 0 under ω and β had weight 1 under ω . So we could have an infinite # of tuples of weight 2: e.g. $(\beta, \beta), (\beta, \beta, \gamma), (\beta, \beta, \gamma, \gamma), \dots$

How does $\Phi_S^\omega(x)$ relate to $\Phi_{S^*}^{\omega^*}(x)$? Note that

$$\begin{aligned} \Phi_{S=\{0,1\}}^\omega(x) &= \sum_{\alpha \in S} x^{\omega(\alpha)} = x^{\omega(0)} + x^{\omega(1)} = 2x \\ \Phi_{S^*}^{\omega^*}(x) &= \Phi_{\{(0),(0),(1),(0,0),(0,1),\dots\}}^{\omega^*}(x) \\ &= \sum_{n \geq 0} (\# \text{ of binary strings of length } n) x^n \\ &= \sum_{n \geq 0} 2^n x^n = \sum_{n \geq 0} (2x)^n \\ &= \frac{1}{1 - 2x} \\ &= \frac{1}{1 - \Phi_S^\omega(x)}. \end{aligned}$$

This is not a coincidence! See the following lemma.

Lemma 2.8 (String Lemma).

Let A be a set with weight function ω such that no elements of A have weight 0. Then,

$$\Phi_{A^*}^{\omega^*}(x) = \frac{1}{1 - \Phi_A^\omega(x)}.$$

Proof. We have

$$\begin{aligned}
\Phi_{A^*}^{\omega^*}(x) &= \Phi_{A^0 \cup A^1 \cup A^2 \cup \dots}^{\omega^*}(x) \\
&= \sum_{n \geq 0} \Phi_{A^n}^{\omega^*}(x) && \text{by Infinite Sum Lemma} \\
&= \sum_{n \geq 0} (\Phi_A^{\omega}(x))^n && \text{by Product Lemma} \\
&= \frac{1}{1 - \Phi_A^{\omega}(x)} && \text{by geometric series.}
\end{aligned}$$

□

2.4 Compositions

Definition 2.8 (Composition).

A **composition** is a finite sequence of positive integers

$$\gamma = (c_1, c_2, \dots, c_k).$$

Each c_i , for $i \in \mathbb{Z}_{>0}$, is called a **part**. The **length** of the composition is the # of parts, i.e. $\text{len}(\gamma) = k$. The **size** of the composition is the sum of the parts, i.e. $|\gamma| = c_1 + c_2 + \dots + c_k$. If γ is a composition of size s , then we say that γ is a composition of s .

Example.

- $(1, 4, 3, 2)$ is a composition of size 10 and with 4 parts.
- $(2, 1, 1)$ is a composition of size 4 and with 3 parts.
- $()$ is a composition of size 0 and with 0 parts (the empty composition).

Theorem 2.9. The generating series for all integer compositions with respect to size is

$$\Phi_{\text{compositions}}^{\text{size}}(x) = \frac{1-x}{1-2x} = 1 + \frac{x}{1-2x}.$$

Example. How many compositions are there of 123?

Proof. By the theorem above, we have

$$\begin{aligned}
 \# \text{ of compositions of } 123 &= [x^{123}] \Phi_{\text{compositions}}^{\text{size}}(x) \\
 &= [x^{123}] \left(\frac{1-x}{1-2x} \right) \\
 &= [x^{123}] (1-x) \sum_{n \geq 0} (2x)^n && \text{by geometric series} \\
 &= [x^{123}] \sum_{n \geq 0} (2x)^n - [x^{123}] \sum_{n \geq 0} x(2x)^n \\
 &= 2^{123} - [x^{122}] \sum_{n \geq 0} (2x)^n \\
 &= 2^{123} - 2^{122} \\
 &= 2^{122}.
 \end{aligned}$$

□

Proof of Theorem 2.9. The generating series for all integer compositions with respect to size is

$$\begin{aligned}
 \Phi_{\text{compositions}}^{\text{size}}(x) &= \Phi_{(\mathbb{Z}_{>0})}^{\text{size}}(x) && \text{since compositions are tuples of positive integers} \\
 &= \frac{1}{1 - \Phi_{\mathbb{Z}_{>0}}^{\text{size}}(x)} && \text{by String Lemma} \\
 &= \frac{1}{1 - \left(\sum_{n \geq 1} x^n \right)} \\
 &= \frac{1}{1 - x \sum_{n \geq 0} x^n} \\
 &= \frac{1}{1 - x \cdot \frac{1}{1-x}} = \frac{1-x}{1-2x}.
 \end{aligned}$$

□

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Theorem 2.10. For $n \in \mathbb{N}$, the # of compositions of size n is

$$\begin{cases} 1, & \text{if } n = 0, \\ 2^{n-1}, & \text{if } n \geq 1. \end{cases}$$

Example. How many compositions are there of 123 where each part is either 1 or 3?

Proof. Let $R = \{1, 3\}$. Then, $\Phi_R^{\text{size}}(x) = x + x^3$. The set of all compositions with each part being 1 or 3 is R^* . So, we want

$$[x^{123}] \Phi_{R^*}^{\text{size}}(x) = [x^{123}] \frac{1}{1 - \Phi_R^{\text{size}}(x)} = [x^{123}] \frac{1}{1 - x - x^3} \quad \text{by String Lemma.}$$

□

Example. Find the generating series for compositions where each part is odd.

Proof. Let $S = \{1, 3, 5, \dots\}$ be the set of odd positive integers. Note that

$$\Phi_S^{\text{size}}(x) = \sum_{n \geq 0} x^{2n+1} = x \sum_{n \geq 0} (x^2)^n = \frac{x}{1 - x^2}.$$

The set of compositions with odd parts is S^* . So, we want

$$\begin{aligned} \Phi_{S^*}^{\text{size}}(x) &= \frac{1}{1 - \Phi_S^{\text{size}}(x)} && \text{by String Lemma} \\ &= \frac{1}{1 - \frac{x}{1 - x^2}} \\ &= \frac{1 - x^2}{1 - x - x^2} \\ &= 1 + \frac{x}{1 - x - x^2}. \end{aligned}$$

□

Example. Find the generating series for compositions where each part is 2 or greater.

Proof. Let $T = \{2, 3, 4, \dots\}$. Note that

$$\Phi_T^{\text{size}}(x) = \sum_{n \geq 2} x^n = x^2 \sum_{n \geq 0} x^n = \frac{x^2}{1 - x}.$$

The set of compositions where each part is ≥ 2 is T^* . By String Lemma, we have

$$\Phi_{T^*}^{\text{size}}(x) = \frac{1}{1 - \Phi_T^{\text{size}}(x)} = \frac{1}{1 - \frac{x^2}{1-x}} = \frac{1-x}{1-x-x^2} = 1 + \frac{x^2}{1-x-x^2}.$$

□

The generating series in the last two examples look rather similar. In fact, there is a strong relation.

Claim. For $n \geq 2$, the # of compositions of n where each part is ≥ 2 is equal to the # of compositions of $n-1$ where each part is odd.

Proof. We want to show that

$$[x^{n-1}] \Phi_{S^*}^{\text{size}}(x) = [x^n] \Phi_{T^*}^{\text{size}}(x).$$

Let's check both sides.

$$\begin{aligned} \text{LHS} &= [x^{n-1}] \left(1 + \frac{x}{1-x-x^2} \right) = [x^{n-1}] 1 + [x^{n-1}] \frac{x}{1-x-x^2} \\ &= 0 + [x^{n-2}] \frac{1}{1-x-x^2} \end{aligned}$$

And,

$$\begin{aligned} \text{RHS} &= [x^n] \left(1 + \frac{x^2}{1-x-x^2} \right) = [x^n] 1 + [x^n] \frac{x^2}{1-x-x^2} \\ &= 0 + [x^{n-2}] \frac{1}{1-x-x^2} \\ &= \text{LHS}. \end{aligned}$$

□

For fixed n , these sets of compositions are finite and of the same size, so can we find a “natural” bijection between them?

Compositions of $n = 7$ with parts ≥ 2	Compositions of $n - 1 = 6$ with odd parts
(7)	(1,5)
(5,2)	(5,1)
(2,5)	(1,1,1,3)
(4,3)	(1,1,3,1)
(3,4)	(1,3,1,1)
(3,2,2)	(3,1,1,1)
(2,3,2)	(3,3)
(2,2,3)	(1,1,1,1,1,1)

Here is a way to construct a bijection.

- (1) Take a composition of 7 with parts ≥ 2 .
- (2) Subtract 1 from the last part to turn into a composition of $n - 1 = 6$.
- (3) For each even part $2k$, split it into $(1, 2k - 1)$.

Compositions of $n = 7$ with parts ≥ 2		Compositions of $n - 1 = 6$ with odd parts
(7)	$\rightarrow$ (6)	$\rightarrow$ (1, 5)
(5,2)	$\rightarrow$ (5, 1)	$\rightarrow$ (5, 1)
(2,5)	$\rightarrow$ (2, 4)	$\rightarrow$ (1, 1, 1, 3)
(4,3)	$\rightarrow$ (4, 2)	$\rightarrow$ (1, 1, 3, 1)
(3,4)	$\rightarrow$ (3, 3)	$\rightarrow$ (3, 3)
(3,2,2)	$\rightarrow$ (3, 2, 1)	$\rightarrow$ (3, 1, 1, 1)
(2,3,2)	$\rightarrow$ (2, 3, 1)	$\rightarrow$ (1, 1, 3, 1)
(2,2,3)	$\rightarrow$ (2, 2, 2)	$\rightarrow$ (1, 1, 1, 1, 1, 1)

Now we have a composition of 6 with only odd parts. However, we still would have to prove that this is a bijection.

3 Binary Strings

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Definition 3.1 (Binary String, Bit).

A **binary string** of length $n \geq 0$ is a finite sequence $\sigma = b_1 b_2 \dots b_n$ where each **bit** $b_i \in \{0, 1\}$.

Example. ϵ (the empty string, which has length 0), 0, 1, 00, 01

Binary strings correspond to elements of the set

$$\{0, 1\}^* = \{(), (0), (1), (0, 0), (0, 1), \dots\}.$$

We already know that the # of binary strings of length n is 2^n , but we can also see this using the String Lemma:

$$\begin{aligned} [x^n] \Phi_{\{0,1\}^*}^{\text{length}}(x) &= [x^n] \frac{1}{1 - \Phi_{\{0,1\}}^{\text{length}}(x)} && \text{(String Lemma)} \\ &= [x^n] \frac{1}{1 - 2x} = 2^n && \text{(Geometric Series)} \end{aligned}$$

Definition 3.2 (Concatenation).

If $\sigma = a_1 a_2 \dots a_m$ and $\tau = b_1 b_2 \dots b_n$ are binary strings, then the **concatenation** $\sigma\tau$ is the binary string $a_1 a_2 \dots a_m b_1 b_2 \dots b_n$.

σ^k denotes the k -fold concatenation of σ with itself, i.e. $\sigma^k = \underbrace{\sigma\sigma\dots\sigma}_{k \text{ times}}$.

Definition 3.3 (Subtring).

We say that σ is a **substring** of τ if there exists (possibly empty) binary strings γ_1, γ_2 such that $\tau = \gamma_1 \sigma \gamma_2$.

Example. Let $\sigma = 011$ and $\tau = 10111$. Note that σ is a substring of τ since

$$\tau = \underbrace{1}_{\gamma_1} \underbrace{011}_{\sigma} \underbrace{1}_{\gamma_2}.$$

Definition 3.4 (Block).

A **block** of a binary string s is a non-empty maximal substring of equal bits.

Example. 00011010000 has 5 blocks: 000, 11, 0, 1, 000.

Example. 000 has 1 block: 000.

Definition 3.5 (Concatenation Product).

If S and T are sets of binary strings and $1 \leq k \in \mathbb{N}$, then

$$ST = \{\sigma\tau : \sigma \in S, \tau \in T\} \quad \text{and} \quad S^k = \underbrace{SS \dots S}_{k \text{ times}}.$$

Example. We have the following concatenation products:

- $\{0, 1\}\{\epsilon, 00, 11\} = \{0, 000, 011, 1, 100, 111\}.$
- $\{00, 01\}^2 = \{0000, 0001, 0100, 0101\}.$
- $\{0, 00\}^2 = \{00, 000, 0000\}.$

Note. From the last example above, the size of the concatenation product ST can sometimes be smaller than $|S| \cdot |T|$.

3.1 Regular Expressions

Definition 3.6 (Regular Expression).

A **regular expression** is defined recursively as any of the following:

- $\epsilon, 0$, or 1 ;
- The expression $R \cup S$ where R and S are regular expressions;
- The expression RS where R and S are regular expressions (with R^k where $k \in \mathbb{N}$);
- The expression R^* where R is a regular expression.

Example. The following are all regular expressions:

- 010 .
- $010 \cup 01$.
- $(010 \cup 01)^*$.
- $(11)(010 \cup 01)^*(\epsilon \cup 0^*)$.
- $(00)^5 \cup (\epsilon \cup 0^*)$.

A regular expression is a formula for **producing** a set of binary strings.

Example.

- $R_1 = \epsilon \cup 0 \cup 1$ produces $\mathcal{R}_1 = \{\epsilon, 0, 1\}$.
- $R_2 = (01)(0 \cup 11)$ produces $\mathcal{R}_2 = \{010, 0111\}$.
- $R_3 = (00 \cup 11)^*$ produces $\mathcal{R}_3 = \{\epsilon, 00, 11, 0000, 0011, 1100, 1111, 000000, \dots\}$.

Definition 3.7 (Production).

We define **production** recursively as follows:

- The regular expressions ϵ , 0 , and 1 produce the sets $\{\epsilon\}$, $\{0\}$, and $\{1\}$ respectively.
- If regular expression R produces the set $\mathcal{R}$ and regular expression S produces the set $\mathcal{S}$, then
 - $R \cup S$ produces $\mathcal{R} \cup \mathcal{S}$.
 - RS produces $\mathcal{RS}$ (concatenation product).
 - R^* produces $\mathcal{R}^* = \bigcup_{k \geq 0} \mathcal{R}^k$ (concatenation powers).

Example.

- 0^* produces $\{\epsilon, 0, 00, 000, \dots\}$.
- $(0 \cup 1)^*$ produces $\{0, 1\}^* = \{\text{all binary strings}\}$.
- $(11)0^*$ produces $\{11, 110, 1100, 11000, \dots\}$.

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- $(00 \cup 000)^*$ produces $\{\epsilon, 00, 000, 0000, 00000, \dots\}$.
- $(0 \cup 00)^*$ produces $\{\epsilon, 0, 00, \dots\}$.

Definition 3.8 (Regular Language).

If $\mathcal{R}$ is a set of binary strings that can be produced by a regular expression R , then we say that $\mathcal{R}$ is a **regular language**. This is also known as a **rational language**.

Example.

- $\{1, 110, 011, \epsilon, 0\}$ is a rational language since it can be produced by $1 \cup 110 \cup 011 \cup \epsilon \cup 0$.
- $\{\underbrace{1010 \dots 10}_{m \text{ times}} \underbrace{00 \dots 0}_{n \text{ times}} : m, n \geq 1\}$ is a rational language and can be produced by $10(10)^*00^*$.
- $\{\epsilon, 01, 0011, 000111, \dots, \underbrace{00 \dots 0}_{n \text{ times}} \underbrace{011 \dots 1}_{n \text{ times}}\}$ is NOT a rational language as no regular expression can produce it.

3.2 Unambiguous Expressions**Example.**

- $(0 \cup 01)(0 \cup 10)$ produces $\{00, 010, 010, 0110\} = \{00, 010, 0110\}$. Note that this regular expression produced 010 in two ways.
- $(0 \cup 1)^*$ produces the set of all binary strings, each string exactly one way.
- $(0 \cup 1 \cup 01)^*$ produces the set of all binary strings but some strings are produced more than once.

Definition 3.9 (Unambiguous Expression).

A regular expression R is **unambiguous** if every string in the language $\mathcal{R}$ produced by R is produced in exactly one way by R . Otherwise, R is called **ambiguous**.

Example. Which of the following are unambiguous? Why?

- 010: unambiguous.
- $(0 \cup 1)^*$: unambiguous.
- $(0 \cup 1)^*(0 \cup 1)^*$: ambiguous. For example, 00 can be produced in two ways as $(00)\epsilon$ or $(0)(0)$.

Example. $0^*(10^*)^*$ is an unambiguous regular expression that produces the set of all binary strings. Note that a string produced by $0^*(10^*)^*$ has the form

$$\underbrace{0 \dots 0}_{m_0} (1 \underbrace{0 \dots 0}_{m_1}) (1 \underbrace{0 \dots 0}_{m_2}) \dots (1 \underbrace{0 \dots 0}_{m_k})$$

for $k \geq 0$ and $m_0, m_1, \dots, m_k \geq 0$.

Every binary string can be written uniquely in this form: k is the number of 1's, and m_i is the number of 0's in the block immediately before the $(i + 1)$ th 1.

Example. $1^*(01^*)^*$ is also an unambiguous regular expression that produces the set of all binary strings.

Example. Give an unambiguous regular expression that produces the set of all binary strings where every 0 is followed by at least three 1's:

$$1^*(01111^*)^*$$

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Remark (Problem Solving Tips: how to justify that a regular expression is unambiguous).

- Argue how to uniquely decompose the string, for example by the positions of the 1's in $0^*(10^*)^*$ (from example above).
- Start from a regular expression A that we already know is unambiguous, and create a new regular expression B for the desired set of strings. Use the Restriction Lemma (below) to prove that B is unambiguous by arguing that every part of B produces a subset of what's produced by the corresponding part of A .

Lemma 3.1 (Restriction Lemma).

Let A_1, A_2, B_1, B_2 be unambiguous regular expressions producing sets S_1, S_2, T_1, T_2 respectively such that $T_1 \subseteq S_1$ and $T_2 \subseteq S_2$. Then,

- (1) If $A_1 \smile A_2$ is unambiguous, then $B_1 \smile B_2$ is also unambiguous.
- (2) If $A_1 A_2$ is unambiguous, then $B_1 B_2$ is also unambiguous.
- (3) If A_1^* is unambiguous, then B_1^* is also unambiguous.

Example. Show that $1^*(01111^*)^*$ is an unambiguous regular expression that produces the set of all binary strings where every 0 is followed by at least three 1's.

Proof. Applying the Restriction Lemma to show this is unambiguous. The sub-expression 01111^* is itself unambiguous and generates a subset of the strings generated by 01^* (and all the other parts of the two expressions are the same). So Restriction Lemma applies. $\square$

3.3 Translation into Generating Series

Example. How many strings of length 10 are there where every block of 1's has even length?

Proof. Let $\mathcal{R}$ be the set of such strings. An unambiguous expression producing $\mathcal{R}$ is

$$R = (0 \smile 11)^*.$$

Since R is unambiguous, there is a bijection between $\{0, 11\}^*$ and $\mathcal{R}$. Then,

$$\begin{aligned} \# \text{ of strings in } \mathcal{R} \text{ of length } 10 &= \# \text{ of strings in } \{0, 11\}^* \text{ of length } 10 \\ &= [x^{10}] \Phi_{\{0,11\}^*}^{\text{length}}(x) \\ &= [x^{10}] \frac{1}{1 - \Phi_{\{0,11\}}^{\text{length}}(x)} \quad (\text{String Lemma}) \\ &= [x^{10}] \frac{1}{1 - (x + x^2)}. \end{aligned}$$

We say that the regular expression R **leads to** the rational function $\frac{1}{1-(x+x^2)}$. $\square$

Definition 3.10 (Leads to).

A regular expression **leads to** a rational function, defined recursively as follows:

- Regular expressions $\epsilon, 0, 1$ lead to formal power series $1, x, x$ respectively.
- If R and S lead to $f(x)$ and $g(x)$, then
 - $R \cup S$ leads to $f(x) + g(x)$.
 - RS leads to $f(x) \cdot g(x)$.
 - R^* leads to $\frac{1}{1-f(x)}$.

Theorem 3.2. let R be a regular expression that unambiguously produces the set $\mathcal{R}$. Suppose that R leads to $f(x)$. Then, f is the generating series for $\mathcal{R}$ with respect to length.

Example.

- Regular expression 1100 leads to $x^1 \cdot x^1 \cdot x^1 \cdot x^1 = x^4$.
- Regular expression $11 \cup 000$ leads to $x^2 + x^3$.
- Regular expression $0^*(11)(11 \cup 000)^*$ leads to $\frac{1}{1-x} \cdot x^2 \cdot \frac{1}{1-(x^2+x^3)}$.

3.4 Block Decomposition

Proposition 3.3. The regular expression $0^*(11^*00^*)^*1^*$ is unambiguous and produces the set of all binary strings. Same for $1^*(00^*11^*)^*0^*$.

Example. Show how each of the above expression can generate the string 00001101000 .

Proof.

- From $0^*(11^*00^*)^*1^*$: $\underbrace{(0000)}_{\text{from } 0^*} \underbrace{(110)}_{\text{from } 11^*00^*} \underbrace{(10000)}_{\text{from } 11^*00^*} \underbrace{()}_{\text{from } 1^*}$.
- From $1^*(00^*11^*)^*0^*$: $\underbrace{()}_{\text{from } 1^*} \underbrace{(000011)}_{\text{from } 00^*11^*} \underbrace{(01)}_{\text{from } 00^*11^*} \underbrace{(0000)}_{\text{from } 0^*}$.

Justification of unambiguity of the block decomposition: the “forced” 1 and 0 inside the parenthesis act as block delimiters, starting a new block each time. □

Example. Give an unambiguous regular expression for the set of all binary strings and not containing 0000 as a substring. Find the generating series.

Proof. Start from the block decomposition $0^*(11^*00^*)^*1^*$, and restrict the blocks of 0's to be length at most 3. Then,

$$R = (\epsilon \cup 0 \cup 00 \cup 000)(11^*(0 \cup 00 \cup 000))^*1^*.$$

The generating series is

$$(1 + x + x^2 + x^3) \cdot \frac{1}{1 - \left(x \frac{1}{1-x} (x + x^2 + x^3)\right)} \cdot \frac{1}{1-x}.$$

□

Example. Find the generating series for all binary strings avoiding 011 as a substring.

Proof. Start from the block decomposition $1^*(00^*11^*)^*0^*$, and restrict to at most one 1 after any block of 0's. Then,

$$T = 1^*(00^*1)^*0^*.$$

This leads to the generating series

$$\Phi_T^{\text{length}}(x) = \frac{1}{1-x} \cdot \frac{1}{1 - \left(x \frac{1}{1-x} x\right)} \cdot \frac{1}{1-x} = \frac{1}{1+x^3}.$$

□

3.5 Recursive Decompositions

We will define something more powerful than regular expressions, called **recursive expressions**, which can reference themselves.

Example. A recursive expression that produces the set of all binary strings is

$$S = \epsilon \cup S(0 \cup 1).$$

This is unambiguous since every non-empty binary string can be uniquely expressed as another binary string with a 0 or 1 appended to the end.

If S leads to generating series $f(x)$, then

$$S = \epsilon \cup S(0 \cup 1) \text{ leads to } f(x) = x^0 + f(x)(x + x) \implies f(x) = \frac{1}{1 - 2x}.$$

A **recursive decomposition** of a set S describes S in terms of itself using the syntax of regular expressions + an extra symbol S representing the set itself.

A recursive decomposition for S is **unambiguous** if each side of the equation produces each string exactly once.

Note. We don't need to prove that the recursive decomposition is unambiguous in this course.

Example.

- $S = 1S1 \cup 0$ produces $\mathcal{S} = \{0, 101, 11011, \dots\}$. A string $\sigma \in \mathcal{S}$ is either 0 or $1\sigma'1$ for some $\sigma' \in \mathcal{S}$. It is unambiguous because σ' is uniquely determined by σ .
- $S = \epsilon \cup 0 \cup 1S1$ produces $\mathcal{S} = \{\epsilon, 0, 11, 101, 1111, 11011, 111111, \dots\}$. This is also unambiguous.
- $T = 0 \cup 00 \cup 0T$ produces $\mathcal{T} = \{0, 00, 000, 0000, \dots\}$ ambiguously.

An unambiguous recursive decomposition leads to an equation involving the generating series for the set in question.

Example. Let $S = \epsilon \cup 0 \cup 1S1$. This leads to

$$\begin{aligned} \Phi_S(x) &= \Phi_\epsilon(x) + \Phi_0(x) + \Phi_1(x) \cdot \Phi_S(x) \cdot \Phi_1(x) \\ &= 1 + x + x^2\Phi_S(x) \\ &= \frac{1+x}{1-x^2} = \frac{1}{1-x}. \end{aligned}$$

Recursive decompositions allow us to produce sets of strings that are not rational languages.

Example. Give a recursive expression for the set $S = \{0^n 1^n : n \in \mathbb{N}\}$.

Proof. Note that $S = 0S1 \cup \epsilon$ produces S . □

3.6 Excluded Substrings

Example. Find the generating series for the set of strings without 100 as a substring.

Proof. We could solve this with a clever regular expression:

$$0^*(1 \cup 10)^*$$

which leads to the generating series

$$\frac{1}{1-x} \cdot \frac{1}{1-(x+x^2)} = \frac{1}{1-2x+x^3}.$$

However, let's try two other ways. We can do it directly using recursive expressions or we can apply the following theorem. □

Theorem 3.4 (Excluded Substring Theorem).

Let $\kappa \in \{0, 1\}^*$ be a non-empty string of length n and let A be the set of binary strings that avoid κ as a substring. Let C be the set of all non-empty suffices γ of κ such that $\kappa\gamma = \eta\kappa$ for some non-empty prefix η of κ . Then,

$$\Phi_A(x) = \frac{1 + \Phi_C(x)}{(1-2x)(1 + \Phi_C(x)) + x^n},$$

Example (Technique 1: Apply the Excluded Substring Theorem).

Apply the Excluded Substring Theorem to the previous example where $\kappa = 100$.

Proof. The non-empty suffices of κ are $S = \{0, 00\}$. The non-empty prefixes of κ are $P = \{1, 10\}$. Is there a suffix $\gamma \in S$ and a prefix $\eta \in P$ s.t. $\kappa\gamma = \eta\kappa$?

Note that

$$\kappa\gamma = \{1000, 10000\}, \quad \eta\kappa = \{1100, 10100\}.$$

The overlaps are: $C = \emptyset$. So, $\Phi_C(x) = 0$.

Then, by the Excluded Substring Theorem,

$$\Phi_A(x) = \frac{1+0}{(1-2x)(1+0)+x^3} = \frac{1}{1-2x+x^3},$$

□

Example (Technique 2: Recursive Expressions).

Use recursive expressions directly to find the generating series for the set of strings without 100 as a substring.

Proof. Let A be the set of strings that avoid 100 as a substring. Let B be the set of strings with exactly 1 occurrence of 100 at the very end.

Claim (1). $B = A100$.

Claim (2). $A \cup B = \epsilon \cup A(0 \cup 1)$.

If Claim (1) is true and unambiguous, then it leads to

$$\Phi_B(x) = \Phi_A(x)\Phi_{100}(x) = \Phi_A(x) \cdot x^3.$$

If Claim (2) is true and unambiguous, then it leads to

$$\Phi_A(x) + \Phi_B(x) = 1 + \Phi_A(x)(x + x).$$

Then, we can solve for $\Phi_A(x)$:

$$\Phi_A(x) + \Phi_A(x)x^3 = 1 + 2x\Phi_A(x)$$

$$\Phi_A(x)(1 + x^3 - 2x) = 1$$

$$\Phi_A(x) = \frac{1}{1-2x+x^3}.$$

□

4 Partial Fractions and Recurrence Relations

Goal: We want to be able to convert between several representatives of a formal power series $F(x) = \sum_{n \geq 0} a_n x^n$:

- (1) An explicit formula for the coefficients: $f_n = [x^n]F(x)$.
- (2) A rational formula for $F(x)$: $F(x) = \frac{p(x)}{q(x)}$ for polynomials $p(x)$ and $q(x)$.
- (3) A recurrence relation for the coefficients of $F(x)$, i.e., a recurrence relation with initial conditions for sequences $f_0, f_1, \dots$

4.1 Rational Function to Explicit Coefficient Formula Using Partial Fractions

Let's try to extract coefficients when the generating series has a multi-term denominator using partial fractions.

Example. Let's consider the following.

$$\begin{aligned}
 [x^7] \frac{1+7x}{1-x-6x^2} &= [x^7] \frac{1+7x}{(1-3x)(1+2x)} \\
 &= [x^7] \left(\frac{-1}{(1+2x)} + \frac{2}{(1-3x)} \right) && \text{(partial fractions)} \\
 &= [x^7] \left(-1 \sum_{n \geq 0} (-2x)^n + 2 \sum_{n \geq 0} (3x)^n \right) && \text{(geometric series)} \\
 &= [x^7] \sum_{n \geq 0} ((-1)(-2)^n + 2 \cdot 3^n) x^n \\
 &= 2^7 + 2 \cdot 3^7.
 \end{aligned}$$

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Theorem 4.1 (Partial Fractions).

Let

$$G(x) = \frac{p(x)}{q(x)} = \frac{p(x)}{(1-\lambda_1 x)^{d_1} (1-\lambda_2 x)^{d_2} \cdots (1-\lambda_s x)^{d_s}},$$

where $\deg(p) < \deg(q)$, the $\lambda_i \in \mathbb{C}$ are distinct, and $d_i \geq 1$. Then, there exists

$C_1^{(1)}, \dots, C_1^{(d_1)}, C_2^{(1)}, \dots, C_2^{(d_2)}, \dots, C_s^{(1)}, \dots, C_s^{(d_s)} \in \mathbb{C}$ such that

$$G(x) = \sum_{i=1}^s \sum_{j=1}^{d_i} \frac{C_i^{(j)}}{(1 - \lambda_i x)^j}.$$

Example. Let $G(x) = \frac{2+5x}{1-3x^2-2x^3}$. Then,

$$\begin{aligned} G(x) &= \frac{2+5x}{(1+x)^2(1-2x)} \\ &= \frac{A}{(1+x)} + \frac{B}{(1+x)^2} + \frac{C}{(1-2x)} \end{aligned} \quad (\text{partial fractions})$$

$$\begin{aligned} 2+5x &= A(1+x)^2 + B(1-2x)(1+x) + C(1-2x) \\ &= (A+B+C) + (-A-2B+2C)x + (-2A+0B-C)x^2. \end{aligned}$$

This leads to a system of equations:

$$\begin{cases} 2 = A + B + C \\ 5 = -A - 2B + 2C \\ 0 = -2A + 0B - C \end{cases}$$

which implies $(A, B, C) = (1, -1, 2)$. So,

$$G(x) = \frac{1}{(1+x)} + \frac{-1}{(1+x)^2} + \frac{2}{(1-2x)}.$$

Now, we can apply the geometric series and negative binomial series to get a single sum:

$$\begin{aligned} G(x) &= \sum_{n \geq 0} (-x)^n - \sum_{n \geq 0} \binom{n+2-1}{2-1} (-x)^n + 2 \sum_{n \geq 0} (2x)^n \\ &= \sum_{n \geq 0} \left((-1)^n - \binom{n+1}{1} (-1)^n + 2 \cdot 2^n \right) x^n \\ &= \sum_{n \geq 0} \left((-1)^n - (n+1)(-1)^n + 2^{n+1} \right) x^n \\ &= \sum_{n \geq 0} \left(2^{n+1} - n(-1)^n \right) x^n. \end{aligned}$$

Also, $[x^n]G(x) = 2^{n+1} - n(-1)^n$.

Remark (Summary).

To get an explicit formula for coefficients of a formal power series given in rational function form, the steps are:

- (1) Factor the denominator.
- (2) Write it as a sum of negative powers of polynomials using partial fractions.
- (3) Solve for e.g. A, B, C by clearing denominators and equating coefficients.
- (4) Use geometric series and negative binomial series to get a single sum.

4.2 Recurrence Relation to Explicit Coefficient Formula

Example. Recall the Fibonacci sequence defined by

$$f_0 = 1, \quad f_1 = 1, \quad f_n = f_{n-1} + f_{n-2} \text{ for } n \geq 2.$$

Let $F(x) = \sum_{n \geq 0} f_n x^n$ be the generating series for the Fibonacci sequence. The recurrence relation is, for $n \geq 2$,

$$\begin{aligned} f_n &= f_{n-1} + f_{n-2} \\ 0 &= f_n - f_{n-1} - f_{n-2} \\ &= [x^n]F(x) - [x^{n-1}]F(x) - [x^{n-2}]F(x) \\ &= [x^n]F(x) - [x^n](xF(x)) - [x^n](x^2F(x)) \\ &= [x^n](1 - x - x^2)F(x). \end{aligned}$$

So for $n \geq 2$, $[x^n](1 - x - x^2)F(x) = 0$. This means that we must have the form

$$(1 - x - x^2)F(x) = C_0 + C_1x + 0x^2 + 0x^3 + \dots \quad \text{for some } C_0, C_1 \in \mathbb{C}.$$

So,

$$F(x) = \frac{C_0 + C_1x}{1 - x - x^2}.$$

To get an explicit form for coefficients of $F(x)$, we first factor the denominator:

$$F(x) = \frac{C_0 + C_1x}{(1 - \alpha x)(1 - \beta x)},$$

where $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$. Using partial fractions, we have

$$F(x) = \frac{A}{(1 - \alpha x)} + \frac{B}{(1 - \beta x)}.$$

Applying the geometric series,

$$F(x) = \sum_{n \geq 0} (A\alpha^n + B\beta^n)x^n.$$

So, $f_n = [x^n]F(x) = A\alpha^n + B\beta^n$. Substituting the initial conditions $f_0 = 1$ and $f_1 = 1$, we get

$$\begin{aligned} f_0 = 1 &= A\alpha^0 + B\beta^0 = A + B, \\ f_1 = 1 &= A\alpha^1 + B\beta^1 = A\frac{1+\sqrt{5}}{2} + B\frac{1-\sqrt{5}}{2}. \end{aligned}$$

Solve the system of 2 linear equations to get

$$A = \frac{5 + \sqrt{5}}{10}, \quad B = -\frac{5 - \sqrt{5}}{10}.$$

We can conclude that

$$f_n = \frac{5 + \sqrt{5}}{10} \left(\frac{1 + \sqrt{5}}{2} \right)^n - \frac{5 - \sqrt{5}}{10} \left(\frac{1 - \sqrt{5}}{2} \right)^n.$$

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Remark (Summary).

To get an explicit formula for coefficients of a formal power series given as a recurrence relation, the steps are:

- (1) Start from the recurrence relation.
- (2) Write a rational expression for the generating series and factor the denominator.

- (3) Apply partial fractions to write expression as a sum.
- (4) Apply geometric and negative binomial theorem to obtain coefficient formula.
- (5) Substitute initial conditions to solve for e.g. A, B .

4.3 Convenient Conversion Theorems

The following theorem shows how to go between a recurrence relation for a generating series and a rational function form for the generating series.

Theorem 4.2 (Recurrence Relation $\leftrightarrow$ Rational Function).

Let $G(x) = \sum_{n \geq 0} g_n x^n$ be a generating series. The following are equivalent:

- (1) The sequence $g_0, g_1, \dots$ satisfies the homogeneous linear recurrence relation

$$g_n + a_1 g_{n-1} + \dots + a_d g_{n-d} = 0$$

for all $n \geq N$ with initial conditions $g_0, g_1, \dots, g_{N-1}$.

- (2) $G(x) = \frac{P(x)}{Q(x)}$ where

$$P(x) = b_0 + b_1 x + b_2 x^2 + \dots + b_{N-1} x^{N-1}$$

$$Q(x) = \underbrace{1 + a_1 x + a_2 x^2 + \dots + a_d x^d}_{\text{auxiliary polynomial}}$$

and

$$b_k = g_k + a_1 g_{k-1} + \dots + a_d g_{k-d}$$

for all $0 \leq k \leq N-1$ with the convention that $g_n = 0$ for all $n < 0$.

The following theorem shows how to go from a recurrence relation to an explicit formula for each coefficient.

Theorem 4.3 (Recurrence Relation $\rightarrow$ Explicit Coefficient Formula).

Suppose $g_0, g_1, \dots$ is a sequence satisfying recurrence relation

$$g_n + a_1 g_{n-1} + \dots + a_d g_{n-d} = 0.$$

Factor the auxiliary polynomial to obtain the “inverse roots” $\lambda_i \in \mathbb{C}$ (distinct) and their multiplicities:

$$1 + a_1x + a_2x^2 + \cdots + a_dx^d = (1 - \lambda_1x)^{d_1}(1 - \lambda_2x)^{d_2} \cdots (1 - \lambda_sx)^{d_s}.$$

Then,

$$g_n = p_1(n)\lambda_1^n + p_2(n)\lambda_2^n + \cdots + p_s(n)\lambda_s^n,$$

where each p_i is a polynomial of degree less than d_i .

Example. Suppose g_n is a sequence given by $(g_0, g_1, g_2) = (0, -5, -1)$ and $g_n = 3g_{n-2} - 2g_{n-3}$ for $n \geq 3$. Give a formula for g_n as a function of n .

Proof. Write the recurrence relation as $g_n - 3g_{n-2} + 2g_{n-3} = 0$. The auxiliary polynomial of the recurrence relation is

$$1 - 3x^2 + 2x^3 = (1 - x)^2(1 + 2x).$$

So by Theorem 4.3, we can write

$$g_n = (An + B)1^n + C(-2)^n.$$

Substituting in our initial conditions $g_0 = 0$, $g_1 = -5$, $g_2 = -1$, and solving for A, B, C yields $(A, B, C) = (-2, -1, 1)$. Thus,

$$g_n = -2n - 1 + (-2)^n.$$

□

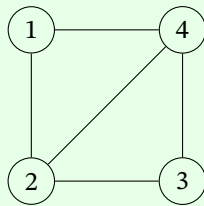
5 Introduction to Graph Theory

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Definition 5.1 (Graph, Vertices, Edges).

A **graph** G consists of a finite non-empty set $V(G)$ of objects, called **vertices**, and a set $E(G)$ of **edges**, which are unordered pairs of distinct vertices.

Example. Let $V(G) = \{1, 2, 3, 4\}$ and $E(G) = \{\{1, 2\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}$. Then, G can be represented as follows.



A graph is the relationship: the set of vertices and edges. But, there are many representations of the same graph.

Definition 5.2 (Adjacent).

Two vertices u and v are **adjacent** if $\{u, v\}$ is an edge.

Example. In the above graph, 1 and 2 are adjacent, but 1 and 3 are not adjacent.

Definition 5.3 (Incident).

If $e = \{u, v\}$ is an edge, then we say that edge e is **incident** with vertices u and v .

Example. In the above graph, $e = \{1, 2\}$ is incident with vertices 1 and 2.

Definition 5.4 (Neighbour).

The vertices adjacent to a vertex v are called **neighbours** of v . The set of neighbours of v is denoted $N(v)$ (sometimes $N_G(v)$).

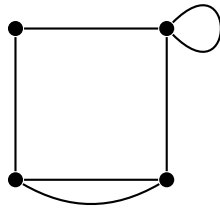
Example. In the above graph, the neighbours of vertex 2 are 1, 3, and 4. So, $N(2) = \{1, 3, 4\}$.

Sometimes we use the shorthand $e = uv$ to denote the edge $e = \{u, v\}$. Edges in this course are unordered (i.e. undirected), so $e = uv = vu$.

Definition 5.5 (Simple Graph).

A **simple graph** is a graph that has no loops (i.e. edges with the same vertex at both ends) and no multiple edges between the same two vertices.

A loop and multiple edges look like this:

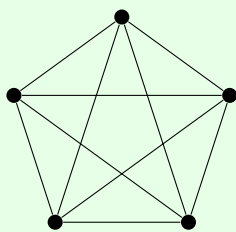


In this course, we only deal with simple graphs.

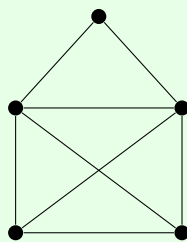
Definition 5.6 (Planar Graph).

A graph is **planar** if it can be drawn without its edges crossing.

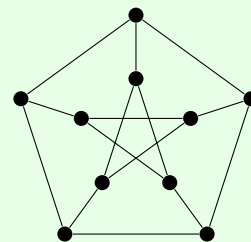
Example.



not planar

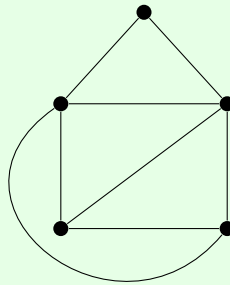


planar



not planar

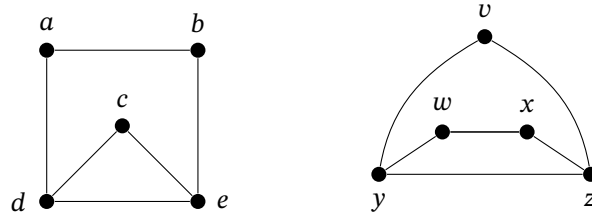
Note that the drawing for the middle graph is not planar, but there is a planar drawing of it:



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5.1 Isomorphism

Are these two graphs the same?



Answer: Not exactly the same: different labels on the vertices.

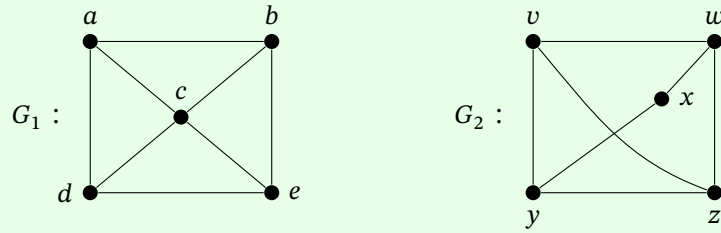
Definition 5.7 (Isomorphism).

Two graphs G_1, G_2 are **isomorphic** if $\exists$ a bijection $f : V(G_1) \rightarrow V(G_2)$ such that u, v are adjacent in $G_1 \iff f(u), f(v)$ are adjacent in G_2 . Such a bijection f is called an **isomorphism**.

To show that two graphs are isomorphic: write down the bijection and argue it preserves all adjacency relationships.

To show that two graphs are not isomorphic: find some property that holds in one graph but not the other, such as number of vertices, number of edges, connected or not, planar or not, etc.

Example.



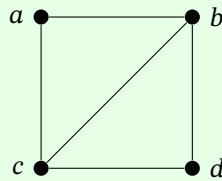
These are not isomorphic because in G_1 , there is a vertex c that is adjacent to 4 vertices, but in G_2 , no vertex is adjacent to 4 vertices.

5.2 Degree

Definition 5.8 (Degree).

The **degree** of a vertex v in a graph G is the number of edges incident with v , and is denoted $\deg(v)$ or $\deg_G(v)$. Equivalently, degree is the size of the neighbourhood: $\deg(v) = |N(v)|$.

Example.



In this graph, $\deg(a) = 2$, $\deg(b) = 3$, $\deg(c) = 3$, and $\deg(d) = 2$. Note that the sum of all degrees is $2 + 3 + 3 + 2 = 10$, which is twice the number of edges.

Theorem 5.1 (Handshaking Lemma).

For every graph G , we have

$$\sum_{v \in V(G)} \deg(v) = 2|E(G)|.$$

Proof. Every edge $e = uv$ has 2 ends, so when we sum the degrees of the vertices, we are counting each edge twice, once at each end (once in $\deg(u)$ and once in $\deg(v)$). $\square$

Corollary 5.2. In any graph, the number of vertices of odd degree is even.

Proof. Let O and E be the sets of vertices with odd and even degree, respectively. Then,

$$\sum_{v \in V(G)} \deg(v) = \sum_{v \in O} \deg(v) + \sum_{v \in E} \deg(v).$$

By the Handshaking Lemma, the LHS is even. By definition of E , $\deg(v)$ is even for all $v \in E$, so $\sum_{v \in E} \deg(v)$ is even. By definition of O , $\deg(v)$ is odd for all $v \in O$. But they sum to an even number, so there must be an even number of odd terms in the sum. Thus, $|O|$ is even. $\square$

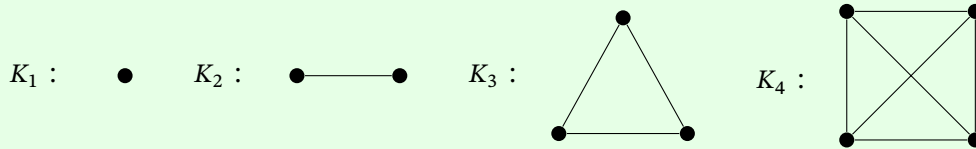
Definition 5.9 (k -regular Graph).

A **k -regular graph** is a graph in which all vertices have degree k .

Definition 5.10 (Complete Graph).

A **complete graph** is a graph where every vertex is adjacent to every other vertex. The complete graph on n vertices is denoted K_n . Equivalently, K_n is an $(n - 1)$ -regular graph.

Example. Here are pictures of K_1, K_2, K_3 , and K_4 .



Question: How many edges does K_n have?

Answer: $\frac{n(n-1)}{2} = \binom{n}{2}$ edges.

Question: How many edges does a k -regular graph with n vertices have?

Answer: By the Handshaking Lemma,

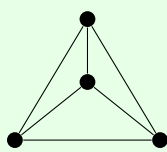
$$\sum_{v \in V(G)} \deg(v) = nk = 2|E(G)| \implies |E(G)| = \frac{nk}{2}.$$

5.3 Bipartite Graphs

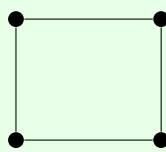
Definition 5.11 (Bipartite Graph).

A graph G is **bipartite** if its vertex set can be partitioned into two disjoint sets A, B such that $V(G) = A \cup B$ and every edge in G has one endpoint in A and one endpoint in B .

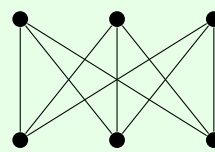
Example. Here are some graphs:



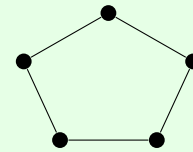
not bipartite



bipartite



bipartite

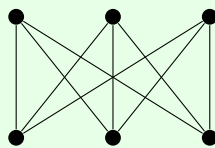


not bipartite

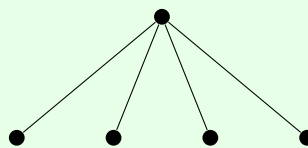
Definition 5.12 (Complete Bipartite Graph).

For positive integers m, n , the **complete bipartite graph** $K_{m,n}$ is the graph with bipartition A, B where $|A| = m, |B| = n$, containing all possible edges joining vertices in A with vertices in B .

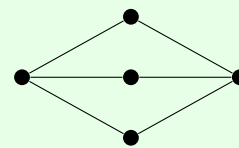
Example. $K_{3,3}$, $K_{1,4}$, and $K_{2,3}$ are shown below.



$K_{3,3}$



$K_{1,4}$



$K_{2,3}$

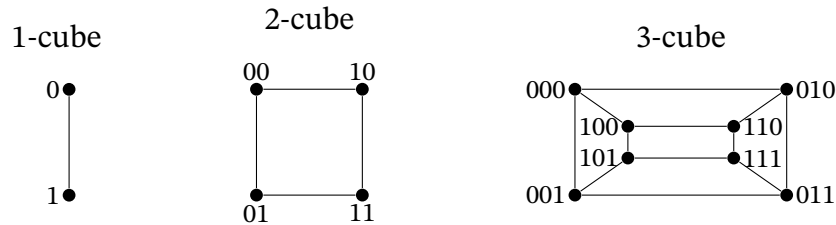
Question: How many edges are in $K_{m,n}$?

Answer: mn edges.

Definition 5.13 (n -Cube or Hypercube).

For $n \geq 0$, the **n -cube** (or **hypercube**) is the graph whose vertex set is the set of all binary

strings of length n , and two vertices are adjacent if their binary strings differ in exactly one position.



Characteristics of the n -cube:

- Number of vertices: 2^n , i.e. the number of binary strings of length n .
- Regularity: n -regular, since for each string s , we can get a neighbourhood of s by changing one of the n positions of s .
- Number of edges: by the Handshaking Lemma,

$$E(G) = \frac{1}{2} \sum_{v \in V(G)} \deg(v) = \frac{1}{2} \cdot 2^n \cdot n = n2^{n-1}.$$

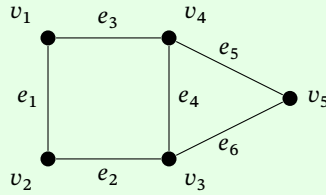
- Bipartiteness: the n -cube is bipartite. To see this, let A be the set of binary strings with an odd number of 1's and let B be the set of binary strings with an even number of 1's. Then, (A, B) partitions the vertex set. Let st be an edge in the graph. By definition, t can be obtained from s by changing one position: either change a 0 to a 1 or a 1 to a 0. In either case, the parity of the number of 1's changes, so the parity of the number of 1's in s and t must be different. Hence, one end of st is in A and the other end is in B . So, the n -cube is bipartite.

We can construct the n -cube recursively:

1. Make 2 copies of the $(n - 1)$ -cube.
2. Pretend 0 in front of all the strings in one copy. Pretend 1 in front of all the strings in the other copy.
3. Joint corresponding copies of the vertices with new edges.

5.4 How to Specify a Graph

Example.



Definition 5.14 (Adjacency Matrix).

The **adjacency matrix** of a graph with vertices $\{v_1, v_2, \dots, v_n\}$ is the $n \times n$ matrix A where

$$A_{ij} = \begin{cases} 1 & \text{if } v_i \text{ and } v_j \text{ are adjacent,} \\ 0 & \text{otherwise.} \end{cases}$$

Example (Continued).

The adjacency matrix for the previous graph is

$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix} \end{matrix}.$$

Note. For simple graphs, A is symmetric and its diagonal entries are all 0.